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C++ Programming Lab Report

1. Aim of the Experiment

The purpose of this laboratory session is to:

Develop understanding of C++ program structure.

Practice writing and executing basic programs.

Use input and output streams effectively.

Work with different variable types.

Perform mathematical operations using operators.

Improve coding style through comments and proper formatting.

Build small applications for problem-solving practice.

2. Background Theory

2.1 Structure of a C++ Program

A standard C++ program contains:

Preprocessor directives (for example, including libraries).

The main() function where execution begins.

Statements inside curly braces { }.

A return statement to end the program.

The program execution always starts from the main() function.

Input and output operations are performed using the standard library.

2.2 Variables and Data Representation

Variables are used to store data in memory. Each variable must have a data type.

Common data types include:

Integer type – stores whole numbers.

Floating-point type – stores decimal values.

Character type – stores single symbols.

Boolean type – stores logical values.

String type – stores text.

Each data type occupies a specific amount of memory.

2.3 Operators in C++

C++ supports several arithmetic operators:

+ for addition

- for subtraction

* for multiplication

/ for division

% for remainder

These operators are commonly used for calculations and problem-solving.

2.4 Programming Style Guidelines

To write clean code:

Use proper indentation.

Add comments to explain logic.

Use meaningful variable names.

Keep the code organized and readable.

Comments help make programs easier to understand and maintain.

3. Practical Implementations

Program 1: Basic Input and Output

Cpp

Copy code

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    string city;
```

```
    int year;
```

```
cout << "Enter your city name: ";
```

```
cin >> city;
```

```
cout << "Enter your birth year: ";
```

```
cin >> year;
```

```
cout << "You live in " << city << " and you were born in " << year << "." << endl;
```

```
return 0;
```

```
}
```

Purpose:

To practice user input and formatted output.

Program 2: Mathematical Expression Evaluator

Cpp

Copy code

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    double a, b;
```

```
    cout << "Enter two numbers: ";
```

```
    cin >> a >> b;
```

```
    cout << "Addition: " << (a + b) << endl;
```

```
    cout << "Subtraction: " << (a - b) << endl;
```

```
    cout << "Multiplication: " << (a * b) << endl;
```

```
    if (b != 0) {
```

```
        cout << "Division: " << (a / b) << endl;
    } else {
        cout << "Division not possible (division by zero)." << endl;
    }

    return 0;
}
```

Purpose:

To apply arithmetic operations and conditional checking.

Program 3: Area of a Rectangle

Cpp

Copy code

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
    double length, width, area;

    cout << "Enter length: ";
    cin >> length;

    cout << "Enter width: ";
    cin >> width;

    area = length * width;

    cout << "Area of rectangle: " << area << endl;

    return 0;
}
```

```
}
```

Purpose:

To perform geometric calculations using variables.

Program 4: Display Variable Sizes

Cpp

Copy code

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    cout << "Size of int: " << sizeof(int) << " bytes" << endl;
```

```
    cout << "Size of double: " << sizeof(double) << " bytes" << endl;
```

```
    cout << "Size of char: " << sizeof(char) << " bytes" << endl;
```

```
    cout << "Size of bool: " << sizeof(bool) << " bytes" << endl;
```

```
    return 0;
```

```
}
```

Purpose:

To understand memory allocation of different data types.

4. Observations

Programs executed successfully.

Input and output operations functioned correctly.

Arithmetic operators produced accurate results.

Memory size of data types was verified.

Conditional statements improved program reliability.

5. Final Conclusion

This laboratory exercise strengthened the understanding of fundamental C++ programming concepts. Through practical implementation, basic syntax, variable declaration, input/output handling, and arithmetic operations were practiced. The tasks enhanced logical thinking and programming confidence, forming a strong foundation for advanced topics such as loops, functions, and object-oriented programming.